```
!pip install transformers datasets scikit-learn torch
```

```
Requirement already satisfied: transformers in /usr/local/lib/python3.12/dist-p
Requirement already satisfied: datasets in /usr/local/lib/python3.12/dist-packa
Requirement already satisfied: scikit-learn in /usr/local/lib/python3.12/dist-p
Requirement already satisfied: torch in /usr/local/lib/python3.12/dist-packages
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Requirement already satisfied: huggingface-hub<1.0,>=0.34.0 in /usr/local/lib/p
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Requirement already satisfied: tokenizers<0.22,>=0.21 in /usr/local/lib/python3
Requirement already satisfied: safetensors>=0.4.3 in /usr/local/lib/python3.12/
Requirement already satisfied: tqdm>=4.27 in /usr/local/lib/python3.12/dist-pac
Requirement already satisfied: pyarrow>=15.0.0 in /usr/local/lib/python3.12/dis
Requirement already satisfied: dill<0.3.9,>=0.3.0 in /usr/local/lib/python3.12/
Requirement already satisfied: pandas in /usr/local/lib/python3.12/dist-package
Requirement already satisfied: xxhash in /usr/local/lib/python3.12/dist-package
Requirement already satisfied: multiprocess<0.70.17 in /usr/local/lib/python3.1
Requirement already satisfied: fsspec<=2025.3.0,>=2023.1.0 in /usr/local/lib/py
Requirement already satisfied: scipy>=1.6.0 in /usr/local/lib/python3.12/dist-p
Requirement already satisfied: joblib>=1.2.0 in /usr/local/lib/python3.12/dist-
Requirement already satisfied: threadpoolctl>=3.1.0 in /usr/local/lib/python3.1
Requirement already satisfied: typing-extensions>=4.10.0 in /usr/local/lib/pyth
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Requirement already satisfied: sympy>=1.13.3 in /usr/local/lib/python3.12/dist-
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Requirement already satisfied: aiohttp!=4.0.0a0,!=4.0.0a1 in /usr/local/lib/pyt
Requirement already satisfied: hf-xet<2.0.0,>=1.1.3 in /usr/local/lib/python3.1
Requirement already satisfied: charset-normalizer<4,>=2 in /usr/local/lib/pytho
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-p
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.12/
Requirement already satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/python3.12/
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/dis
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-p
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist
Requirement already satisfied: aiohappyeyeballs>=2.5.0 in /usr/local/lib/pytho
Requirement already satisfied: aiosignal>=1.4.0 in /usr/local/lib/python3.12/di
Requirement already satisfied: attrs>=17.3.0 in /usr/local/lib/python3.12/dist-
Requirement already satisfied: frozenlist>=1.1.1 in /usr/local/lib/python3.12/d
```

```
import pandas as pd
import numpy as np
from sklearn.model_selection import train_test_split
from sklearn.metrics import classification_report, confusion_matrix, accuracy_score

import torch
from torch.utils.data import DataLoader
from transformers import BertTokenizer, BertForSequenceClassification, Trainer, Trai
```

```
df = pd.read_csv("/content/cleaned_customer_support_english.csv")
df.head(5)
```

	cleaned_text	type	priority	queue
0	account disruption dear customer support teamn...	Incident	high	Technical Support
1	query smart home system integration features d...	Request	medium	Returns and Exchanges
2	inquiry regarding invoice details dear custome...	Request	low	Billing and Payments
[illegible]	question marketing agency software [illegible]	[illegible]	[illegible]	[illegible]

Next steps: Generate code with df | View recommended plots | New interactive sheet

```
df['type'].value_counts()
```

type	count
Incident	4717
Request	3320
Problem	2413
Change	1226

dtype: int64

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 11676 entries, 0 to 11675
Data columns (total 5 columns):
 #   Column        Non-Null Count  Dtype
---  ------        --------------  -----
 0   cleaned_text  11676 non-null  object
 1   type          11676 non-null  object
 2   priority      11676 non-null  object
 3   queue         11676 non-null  object
 4   label         11676 non-null  int64
dtypes: int64(1), object(4)
memory usage: 456.2+ KB
```

```
from sklearn.preprocessing import LabelEncoder
label_encoder = LabelEncoder()
df['label'] = label_encoder.fit_transform(df['type'])
```

```
df.head(5)
```

	cleaned_text	type	priority	queue	label
0	account disruption dear customer support teamn...	Incident	high	Technical Support	1
1	query smart home system integration features d...	Request	medium	Returns and Exchanges	3
2	inquiry regarding invoice details dear custome...	Request	low	Billing and Payments	3
[illegible]	question marketing agency software [illegible]	[illegible]	[illegible]	Sales and Pre-	[illegible]

Next steps: Generate code with df | View recommended plots | New interactive sheet

```
print("Encoded Labels:\n", df[['type','label']].drop_duplicates())
```

```
Encoded Labels:
          type  label
0   Incident      1
1    Request      3
3    Problem      2
15    Change      0
```

```
train_texts, test_texts, train_labels, test_labels = train_test_split(
    df['cleaned_text'], df['label'], test_size=0.2, random_state=42, stratify=df['la
)
```

```
tokenizer = BertTokenizer.from_pretrained('bert-base-uncased')

train_encodings = tokenizer(list(train_texts), truncation=True, padding=True, max_le
test_encodings = tokenizer(list(test_texts), truncation=True, padding=True, max_leng

class TicketDataset(torch.utils.data.Dataset):
    def __init__(self, encodings, labels):
        self.encodings = encodings
        self.labels = labels

    def __len__(self):
        return len(self.labels)

    def __getitem__(self, idx):
        item = {key: torch.tensor(val[idx]) for key, val in self.encodings.items()}
        item['labels'] = torch.tensor(self.labels.iloc[idx])
        return item

train_dataset = TicketDataset(train_encodings, train_labels)
test_dataset = TicketDataset(test_encodings, test_labels)
```

```
/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:94: UserW
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access publ
  warnings.warn(
```

tokenizer_config.json: 100% 48.0/48.0 [00:00<00:00, 2.96kB/s]

vocab.txt: 100% 232k/232k [00:00<00:00, 9.43MB/s]

tokenizer.json: 100% 466k/466k [00:00<00:00, 6.71MB/s]

config.json: 100% 570/570 [00:00<00:00, 63.9kB/s]

```
model = BertForSequenceClassification.from_pretrained('bert-base-uncased', num_label
```

model.safetensors: 100% 440M/440M [00:07<00:00, 89.2MB/s]

```
Some weights of BertForSequenceClassification were not initialized from the mode
You should probably TRAIN this model on a down-stream task to be able to use it
```

```
!pip install -U transformers
```

```
Requirement already satisfied: transformers in /usr/local/lib/python3.12/dist-pa
Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packag
Requirement already satisfied: huggingface-hub<1.0,>=0.34.0 in /usr/local/lib/py
```

```
Requirement already satisfied: numpy>=1.17 in /usr/local/lib/python3.12/dist-pac
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist
Requirement already satisfied: pyyaml>=5.1 in /usr/local/lib/python3.12/dist-pac
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Requirement already satisfied: fsspec>=2023.5.0 in /usr/local/lib/python3.12/dis
Requirement already satisfied: typing-extensions>=3.7.4.3 in /usr/local/lib/pyth
Requirement already satisfied: hf-xet<2.0.0,>=1.1.3 in /usr/local/lib/python3.12
Requirement already satisfied: charset-normalizer<4,>=2 in /usr/local/lib/python
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-pa
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/d
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.12/d
```

```
training_args = TrainingArguments(
    output_dir='./results',
    do_eval=True,
    learning_rate=2e-5,
    per_device_train_batch_size=16,
    per_device_eval_batch_size=16,
    num_train_epochs=3,
    weight_decay=0.01,
    logging_dir='./logs',
    report_to=None   # disables wandb
)
```

```
trainer = Trainer(
    model=model,
    args=training_args,
    train_dataset=train_dataset,
    eval_dataset=test_dataset,
    tokenizer=tokenizer,
)
```

```
Using the `WANDB_DISABLED` environment variable is deprecated and will be remove
/tmp/ipython-input-3876680192.py:14: FutureWarning: `tokenizer` is deprecated an
  trainer = Trainer(
```

```
trainer.train()
trainer.evaluate()
```

[1752/1752 11:18, Epoch 3/3]

Step	Training Loss
500	0.268600
1000	0.247100
1500	0.158300

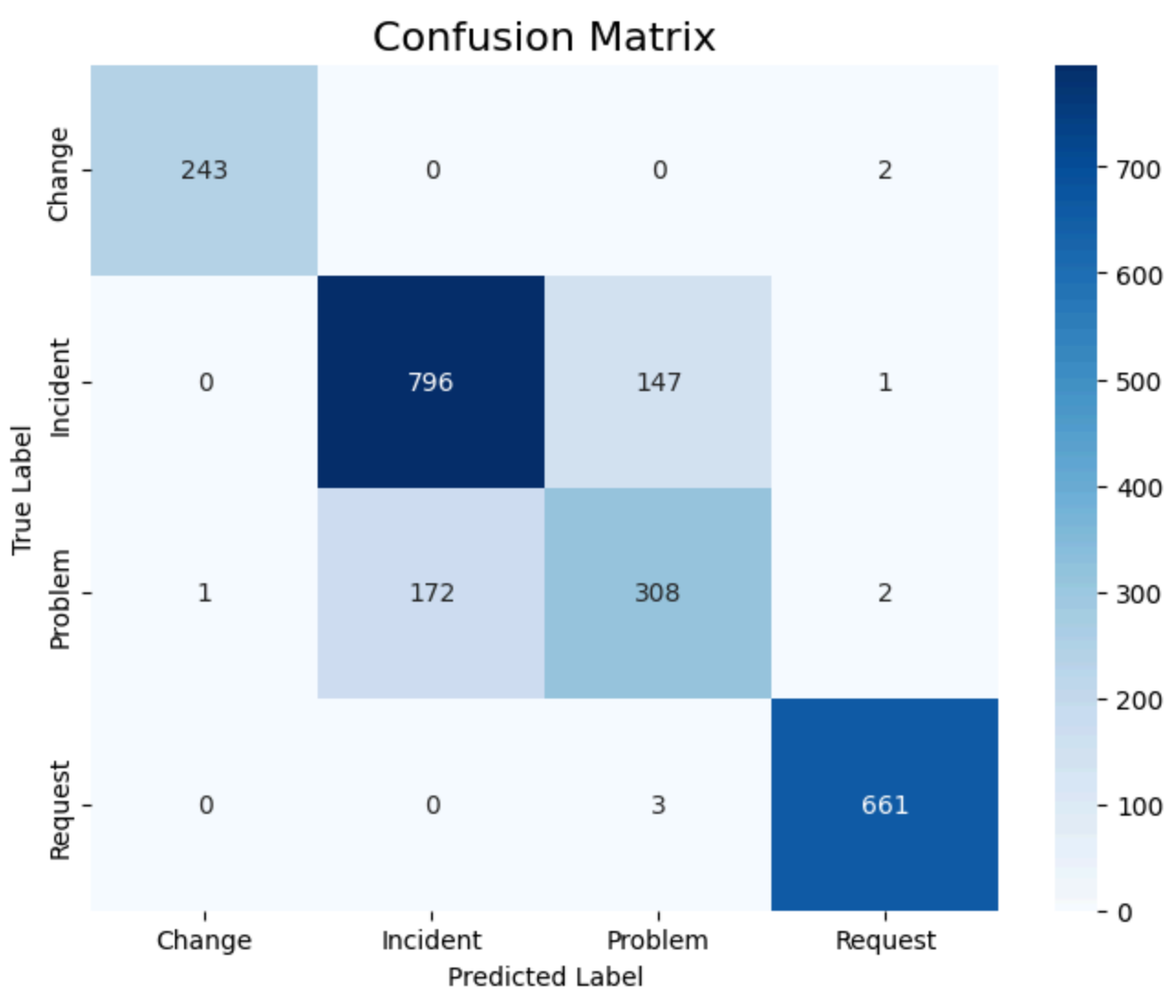
[146/146 00:13]

```
{'eval_loss': 0.513355016708374,
 'eval_runtime': 13.87,
 'eval_samples_per_second': 168.421,
 'eval_steps_per_second': 10.526,
 'epoch': 3.0}
```

```
predictions = trainer.predict(test_dataset)
y_pred = np.argmax(predictions.predictions, axis=1)
y_true = test_labels

print("Accuracy:", accuracy_score(y_true, y_pred))
print("\nClassification Report:\n", classification_report(y_true, y_pred, target_nam
```

```
Accuracy: 0.8595890410958904

Classification Report:
               precision    recall  f1-score   support

      Change       1.00      0.99      0.99       245
    Incident       0.82      0.84      0.83       944
     Problem       0.67      0.64      0.65       483
     Request       0.99      1.00      0.99       664

    accuracy                           0.86      2336
   macro avg       0.87      0.87      0.87      2336
weighted avg       0.86      0.86      0.86      2336
```

```
import matplotlib.pyplot as plt
import seaborn as sns
cm = confusion_matrix(y_true, y_pred)
labels = label_encoder.classes_

plt.figure(figsize=(8,6))
sns.heatmap(cm, annot=True, fmt="d", cmap="Blues", xticklabels=labels, yticklabels=l
plt.title("Confusion Matrix", fontsize=16)
plt.xlabel("Predicted Label")
plt.ylabel("True Label")
plt.show()
```




```
trainer.save_model("final_model")
tokenizer.save_pretrained("final_model")
```

```
('final_model/tokenizer_config.json',
 'final_model/special_tokens_map.json',
 'final_model/vocab.txt',
 'final_model/added_tokens.json')
```

```
import json

label_map = {
    0: "Change",
    1: "Incident",
    2: "Problem",
    3: "Request"
}

with open("label_map.json", "w") as f:
    json.dump(label_map, f)

from google.colab import files
files.download("label_map.json")
```

```
from google.colab import drive
drive.mount('/content/drive')

model.save_pretrained("/content/drive/MyDrive/final_model")
tokenizer.save_pretrained("/content/drive/MyDrive/final_model")
```

```
Mounted at /content/drive
('/content/drive/MyDrive/final_model/tokenizer_config.json',
 '/content/drive/MyDrive/final_model/special_tokens_map.json',
 '/content/drive/MyDrive/final_model/vocab.txt',
 '/content/drive/MyDrive/final_model/added_tokens.json')
```